Homework 2: Psychophysics Mini Study

115-1 Human Perceptual-Centric Vision and Multimodal Representation Learning

Schedule: Released on **October 13, 2026 (Week 6)**; related psychophysics lecture on **October 20, 2026 (Week 7)**; final submission due before class on **November 3, 2026 (Week 9)**. Students will have approximately **three weeks** to complete the assignment.

1 Objective and Learning Goals

The goal of this assignment is to learn how to translate a perceptual question into a small but controlled human experiment. Students will design stimuli, select an appropriate psychophysical or subjective evaluation method, collect pilot data, and perform basic quantitative analysis.

The emphasis is **not** on obtaining a statistically significant result. Instead, the assignment focuses on whether the research question, experimental manipulation, human judgment, and analysis are logically connected.

By completing this assignment, students should learn how to define measurable variables, control confounding factors, design a reproducible perceptual experiment, and interpret variability in human responses. Students should also recognize that human perceptual data are not simply "ground-truth labels"; responses depend on the stimuli, task, protocol, participants, and analysis procedure.

Perceptual Question → Controlled Stimuli → Human Judgment → Quantitative Analysis → Interpretation

The central question to keep in mind is:

Does the experiment actually measure the perceptual phenomenon that you claim it measures?

2 Task and Methods

Design and conduct a small **pilot psychophysics or subjective perception study** investigating one clearly controlled perceptual factor.

Possible topics include contrast, blur, noise, compression artifacts, spatial frequency, color distortion, chroma subsampling, frame rate, motion blur, image quality, visual saliency, or other perception-related factors discussed in class.

Choose **one primary evaluation method** appropriate for the research question:

2AFC / Pairwise Comparison: Participants compare two alternatives and select one according to a specified criterion. This method is suitable for discrimination, preference, or relative quality judgments.

MOS / Rating: Participants assign a numerical rating, such as a 1–5 or 1–7 scale. This method is suitable for perceived quality, intensity, naturalness, or similar subjective attributes.

Staircase / Threshold Estimation: Stimulus intensity is adjusted according to previous responses to estimate a perceptual threshold or JND.

Other psychophysical procedures may be used if clearly justified.

3 Team and Participants

This homework should be completed in teams of **2–3 students**. Each team will design, implement, conduct, and analyze one pilot experiment.

Each team must collect at least **5 valid participant responses**. Team members should not be counted as participants in their own experiment, but students are encouraged to participate in experiments conducted by other teams in the class.

This is a small pilot study intended to evaluate whether the experimental design produces interpretable measurements rather than to support strong population-level conclusions. Participation should be voluntary, and unnecessary personally identifiable or sensitive information should not be collected.

A short **Contribution Statement** describing each team member's role should be included at the end of the report.

4 Minimum Experimental Requirements

The experiment should manipulate at least **one clearly defined independent variable**, such as blur level, contrast, compression strength, spatial frequency, chroma resolution, frame rate, or distortion location.

At least **3 meaningful stimulus conditions or levels** are required, unless a staircase procedure generates stimulus levels adaptively.

Only the intended perceptual factor should be systematically manipulated. Other stimulus properties should remain as constant as reasonably possible. For example, an experiment on blur should not unintentionally change image brightness, crop size, resolution, or compression level.

Presentation order should be randomized when appropriate. For pairwise experiments, left/right or first/second presentation order should also be randomized or counterbalanced.

Participant instructions should clearly define the judgment being requested, and the same criterion should be used for all participants.

At least one simple quality-control procedure should be included, such as checking incomplete trials, detecting invalid responses, verifying that responses were recorded correctly, or examining unusually inconsistent response patterns.

Do not over-engineer the experimental interface. A simple Python script, Jupyter notebook, local webpage, or minimal interface is sufficient as long as stimuli are presented correctly, responses are recorded accurately, and the required randomization or counterbalancing is implemented. A production-quality interface does not receive additional credit.

5 Required Content

The report should contain the following components.

1. Research Question and Hypothesis State the perceptual question and provide a concrete, testable prediction. Specify what variable is manipulated, what human response is measured, and how you expect the response to change across conditions.

2. Stimulus and Experimental Design Describe how the stimuli were generated or selected, what variable was manipulated, what factors were controlled, and why the selected psychophysical

method is appropriate.

Include at least one simple diagram, stimulus example, or experimental pipeline that allows the reader to understand the study design quickly.

3. Protocol Describe the participant task, number of trials, presentation order, randomization or counterbalancing strategy, response procedure, and quality-control method. The description should contain enough information for another student to approximately reproduce the experiment.

4. Results and Analysis Present the pilot results using at least **one quantitative figure or table**.

For a **2AFC or pairwise comparison** study, report the preference or selection rate for each condition and include a confidence interval when appropriate.

For a **MOS or rating** study, report the mean or median rating for each condition together with variability or confidence intervals.

For a **staircase or threshold** study, visualize the staircase trajectory and estimate the perceptual threshold or JND.

More advanced analyses, such as psychometric-function fitting, d', bootstrap confidence intervals, inter-rater reliability, or statistical tests, are welcome but are not required unless needed to answer the research question.

5. Interpretation and Limitations Explain whether the observed results are consistent with the hypothesis and what perceptual mechanism might explain the pattern. Discuss at least one limitation of the current pilot study and one change that should be made before conducting a larger experiment.

Because the study contains only a small number of participants, avoid making strong population-level claims.

6 Submission

Submit the following materials:

(1) Report: A concise PDF report describing the research question, experimental design, protocol, results, and interpretation.

(2) Experimental Materials: The stimuli and the code, notebook, webpage, or other materials used to conduct the experiment.

The recommended report length is **3 pages**, with an absolute maximum of **4 pages**, excluding references and the Contribution Statement.

Students should prioritize a clear experimental design and interpretable figures over lengthy background discussion.

This pilot study may later be extended as part of the final project by adding participants, conditions, perceptual metrics, or machine-learning models.

All submissions must follow the course-wide *Academic Integrity and Generative AI Policy*.

7 Grading

Component	Points
Research question and testable hypothesis	15
Stimulus design and control of variables	20
Experimental protocol and method selection	20
Pilot implementation and data collection	10
Quantitative analysis and visualization	20
Interpretation, limitations, and proposed improvements	10
Clarity and reproducibility of the report	5
Total	**100**

The strongest submissions will demonstrate a clear connection between the research question, stimulus manipulation, human measurement, and analysis. A technically simple experiment with well-controlled variables and a thoughtful interpretation is preferable to a complicated experiment with unclear controls.